

Extracting Cognitive State from Fidgeting Behavior via a Soft Tangible Interface

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Abstract

We investigate whether spontaneous fidgeting can serve as a behavioral signal of cognitive load under time pressure. Data were analyzed from adults with ADHD in a study designed to assess the relation between fidgeting and cognitive regulation. We analyzed tactile interaction traces captured via a sensor-enabled soft fidget ball during the Paced Auditory Serial Addition Test (PASAT), a task that systematically varies time pressure and cognitive demand. We apply supervised machine learning models to fidgeting signals to (1) detect levels of task-induced time pressure and (2) predict task performance outcomes. Our results show that tactile fidgeting patterns reliably differentiate levels of time pressure and are correlated with performance accuracy during PASAT. These patterns suggest that soft tangible interfaces may offer an unobtrusive behavioral channel for sensing cognitive load and informing adaptive systems.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Cognitive computing*; Interaction devices; • **Computing methodologies** → *Machine learning*.

Keywords

ADHD, fidgeting behavior, tactile interaction, cognitive load, affective computing

ACM Reference Format:

Nahid Nasiri, Katherine Isbister, Gabriel H Elkaim, Peter S Cottrell, Julie B Schweitzer, Jared Borden, and Daniel Goodman Shapiro. 2026. Extracting Cognitive State from Fidgeting Behavior via a Soft Tangible Interface. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798867>

1 Introduction

Fidgeting is typically regarded as an unconscious behavior, often interpreted as a sign of distraction or restlessness. However, emerging work in both psychology and human-computer interaction (HCI) suggests that fidgeting can play a constructive role in self-regulation, particularly in neurodivergent populations such as individuals with ADHD [27]. Despite growing interest, fidgeting remains underexplored as a measurable input modality in cognitive state sensing. This is particularly relevant for individuals with ADHD, whose challenges with cognitive control and emotion regulation make them ideal candidates for cognition-aware systems. In this work, we do not evaluate whether the device reduces cognitive load. Rather, we investigate whether spontaneous fidgeting behavior encodes measurable information about task-induced cognitive load.

In this paper, we investigate whether fidgeting, captured through a tangible soft interface, can serve as a passive, unobtrusive signal of users' task-induced cognitive load. Specifically, we employ a previously developed, sensor-enabled fidget ball [26] to analyze tactile interaction data collected during the Paced Auditory Serial Addition Test (PASAT), a cognitively demanding task that systematically varies time pressure. We focus on two questions: 1) Does fidgeting reflect task-induced time pressure? 2) Does fidgeting reflect task performance under cognitive load?



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798867>



Figure 1: Core hardware components used in the first iteration of the ball, displayed from left to right: JW Hol-ee Roller toy covered in a double layer of fabric, Adafruit Feather M0 microcontroller, micro-USB cable, and six 18.5 mm circular capacitive force-sensitive resistors distributed symmetrically around the ball’s surface. Courtesy of [26].

The key technical challenge in realizing such systems is the inference of mental state from observable behavior. This has long been a central goal in cognitive state sensing and human-centered computing, where researchers have primarily relied on modalities such as facial expressions [21], vocal prosody [1], physiological signals [19], and sentiment analysis [25]. In contrast, we investigate a relatively underexplored behavioral channel: spontaneous tactile signals from fidgeting behavior. This is particularly relevant for individuals with Attention-Deficit/Hyperactivity Disorder (ADHD), who frequently use fidgeting as a means of self-regulation [18]. By self-regulation, we refer to the use of motor activity as a mechanism supporting attentional and arousal control in ADHD.

Understanding the cognitive content of tactile signals opens new design opportunities for adaptive systems that passively respond to user cognitive load, especially for neurodivergent populations.

We used the sensor-enabled soft-tangible device in Figure 1 to unobtrusively capture tactile patterns generated during cognitive tasks [8]. Our study applies machine learning techniques to this dataset to test whether spontaneous fidgeting encodes task-induced cognitive load and task performance during PASAT. Our results achieve accuracy levels comparable to recent biosignal identification studies [15], supporting the robustness of fidgeting as an informative behavioral signal. Similar to recent feature-based analyses [15], we employ rigorous validation procedures to ensure reliability. While we do not evaluate cognitive load reduction effects of the device, our findings suggest potential pathways for unobtrusive cognitive load sensing in future adaptive systems. We formulate two hypotheses: H1: Fidgeting patterns differentiate graded levels of task-induced time pressure during PASAT. H2: Fidgeting behavior predicts individual task performance under cognitive load.

We organize the paper as follows. Section 2 situates our contribution within existing work on cognitive computing, tactile interaction, and fidgeting behavior in ADHD. Section 3.1 introduces the data collection context for the dataset, including the sensor-enabled fidget ball and its core hardware components. Section 3.2 describes the experiment tasks that produced the dataset used in our analyses. Section 4 details our machine learning pipeline, including model architecture and training procedures. Section 5 presents two analyses:

detection of task-induced time pressure and prediction of PASAT task performance from fidgeting behavior. We discuss broader implications in Section 6, address study limitations in Section 7, and conclude with future directions in Section 8.

2 Related Work

Research in HCI and psychology has shown that physical movement can support self-regulation [13], particularly in individuals with ADHD, where increased motor activity may compensate for attentional and arousal challenges. Son et al. [27] suggest that aggregate measures of fidgeting behavior correlate with cognitive performance under demanding conditions, while other recent work shows that fidgeting predicts self-assessed measures of cognitive regulation [6, 24]. One interpretation of this relationship is that increased motor activity in ADHD may function as a compensatory mechanism supporting arousal regulation under higher cognitive demand [18]. Under this framework, variations in time pressure may modulate fidgeting intensity as part of adaptive self-regulatory behavior. Our work builds on this perspective by demonstrating that the fine-grained tactile content of fidgeting predicts qualitative and quantitative performance during a time-pressured cognitive task.

Cognitive state analysis has a long interdisciplinary history, connecting psychological theories of cognition and mental workload with computational methods for inferring internal states from observable signals [28, 29]. Early work focused on identifying physiological correlates of internal state, including subtle motor and somatic signals measured during interaction, such as fine-grained muscle activity captured through touch [4]. Subsequent research expanded to automated inference of cognitive load and related states from physiological signals such as facial expression, speech, and biosignals [23] including EEG-based workload classification [16]. Other work demonstrates that touch-based interactions, including pressure and typing dynamics, can encode emotional and cognitive information [11]. Overall, the field has shifted from rule-based approaches with manually engineered features toward data-driven and hybrid machine learning methods capable of modeling complex, nonlinear relationships between behavior and cognitive state [29, 30]. Although facial-expression-based inference is widely studied, recent work has questioned its theoretical and contextual validity for inferring internal emotional states [2], motivating exploration of alternative behavioral channels.

The task of affect recognition from touch is closely related to cognitive recognition and employs similar techniques. For instance, keyboard strokes and touchpad interactions have been shown to reflect emotional states based on features such as typing speed and pressure patterns over time [7, 20, 22]. Horii et al. employed probabilistic neural networks and pressure data to infer six emotional states using simulated input [14], demonstrating the potential of integrating diverse data types in emotion analysis. Additional work by Ghosh et al. applied typing pressure and speed on mobile devices for emotion classification [9, 10], while Shapiro et al. [26] automated the feature extraction process using deep learning methods, supporting the value of touch-based inputs in applications. Building on this line of research, our study applies deep learning

to spontaneous fidgeting behavior as a novel modality for sensing cognitive state.

3 Methods

We analyze tactile fidgeting data collected in a controlled laboratory study with adults diagnosed with ADHD. Below, we summarize the study procedures and participants, followed by the experimental task we employed to induce cognitive load.

3.1 Data Collection Context

Data analyzed in this paper were collected from adults aged 18 to 50 with a documented childhood diagnosis of ADHD and current clinically significant symptoms as part of a larger controlled study. Participants completed tasks while interacting naturally with a sensor-enabled soft fidget ball. The device recorded continuous tactile pressure signals from six embedded sensors, synchronized with task timing to enable alignment between fidgeting behavior and performance measures. Participants were instructed to hold and manipulate the device naturally during task performance, without specific guidance on how or when to interact with it. The study was approved by the Institutional Review Board, and all participants provided informed consent prior to participation.

3.2 Experimental Task

In this paper, we focus on a single cognitively demanding experimental task, the **Paced Auditory Serial Addition Test (PASAT)**, which induces cognitive load by requiring continuous mental addition under time pressure. The PASAT is widely used as a standardized manipulation of cognitive workload and time pressure [12], making it a suitable proxy for graded cognitive load in this study.

The PASAT presents a sequence of single-digit auditory prompts, and participants report the sum of the two most recent digits. The task consists of three phases with inter-prompt intervals of 2.4 seconds (Phase I), 2.0 seconds (Phase II), and 1.6 seconds (Phase III). These progressively shorter intervals increase demands on working memory, attention control, and processing speed [12].

We collected PASAT phase labels and per-phase performance measured as percent correct responses. We use PASAT phase as an operationalized index of graded cognitive load for time-pressure detection, and Phase I percent correct as the target variable for performance prediction.

3.3 Analysis

We employ machine learning to infer cognitive load from spontaneous fidgeting because the relevant tactile features and their relationships to cognitive load are not known *a priori* and are likely nonlinear. Data-driven models can automatically learn complex feature representations and interactions that would be difficult to engineer manually. This approach is consistent with prior work in affective computing and cognitive state inference, where deep learning has been used to decode internal states from physiological and touch-based behavioral signals, as discussed in Section 2.

4 Extracting Cognitive State from Touch via Machine Learning

We employ machine learning to examine the detailed tactile content of the signal obtained from the fidget ball. Rather than relying on hand-crafted features, our models learn patterns directly from raw time-series tactile data, capturing cues related to cognitive load and task performance during PASAT.

4.1 Data

We collected approximately 61 minutes of spontaneous tactile interaction during the PASAT task. The analyzed dataset includes 27 participants in the fidget-ball condition who completed all relevant study procedures and generated sufficient active touch traces while interacting with the fidget ball during this cognitively demanding task. These participants were part of a larger randomized NIH-funded study, in which 49 individuals were included in the primary analysis of the fidget-ball condition.

We captured the participant's fidgeting behavior as the time sequence of signals from six pressure sensors embedded in the fidget ball, preprocessed by thresholding to eliminate low-level noise. Continuous touch traces were segmented into sequences, excluding gaps of 0.5 s or longer. Only sequences that met this activity criterion were retained for analysis.

4.2 Machine Learning Model

We employ supervised deep learning to map fidgeting behavior to PASAT outcomes, as the relevant tactile features and their relationship to cognitive state are unknown *a priori* and likely nonlinear. Our models use 1D convolutional neural networks (CNN) for spatial feature extraction across sensors, followed by fully connected layers for prediction (FCNN). A Long Short-Term Memory (LSTM) layer was included only for PASAT performance prediction to capture temporal dependencies, while time-pressure detection omitted recurrent layers to avoid learning the fixed PASAT prompt rhythm (i.e., a 'metronome' effect).

Models were trained with early stopping using participant-held-out evaluation protocols aligned with each analysis goal. For time-pressure detection, we used leave-one-participant-out validation, averaging accuracy across held-out participants. For PASAT Phase 1 performance prediction, we used participant-based 5-fold cross-validation, with metrics averaged across folds.

5 Results

We present two analyses: detecting levels of task-induced time pressure from fidgeting during PASAT, and predicting PASAT task performance from tactile interaction patterns.

5.1 Detecting Time Pressure Levels from Fidgeting

We evaluated whether fidgeting patterns differentiate levels of experimentally induced time pressure during PASAT. We posed a three-way classification task using the Paced Auditory Serial Addition Test (PASAT), where inter-prompt intervals set cognitive load level: low 2.4 s, moderate 2.0 s, high 1.6 s.

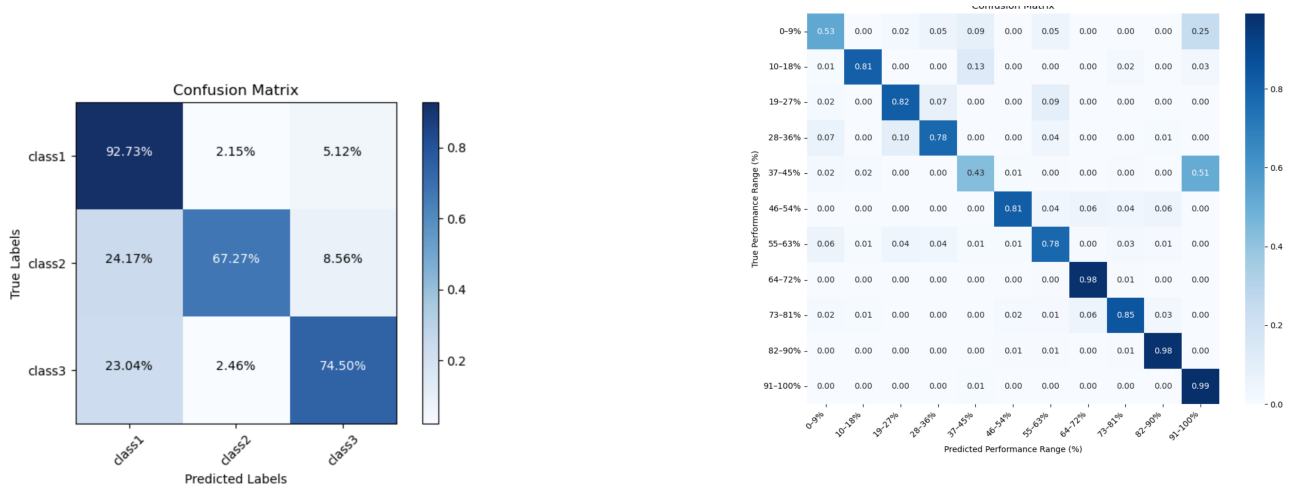


Figure 2: Left: PASAT time-pressure classification from fidgeting across three inter-stimulus intervals. Right: PASAT Phase 1 task performance prediction from tactile interaction patterns.

To prevent a possible metronome effect, we (i) segmented data into uniform 0.5 s windows that never span multiple prompts, and (ii) omitted any LSTM layer so the model cannot learn rhythmic patterns across time. Predictions therefore rely on instantaneous tactile features rather than periodic cues.

We trained a 1D CNN + FCNN architecture against a supervised signal composed of the PASAT phase paired with a 0.5 second window of data collected from the six fidget ball pressure sensors at approximately 40 samples per second. We employed the resulting model for performance prediction. As shown in Figure 2, the model achieved 92% accuracy for low cognitive load, 67% for moderate, and 74% for high, well above the 33% random baseline. Accuracy was computed per 0.5 s segment and averaged across test samples. These results show that fidgeting encodes the qualitative degree of cognitive time-pressured cognitive load.

5.2 Predicting Task Performance from Fidgeting

Architecture: 1D CNN+LSTM+FCNN

Given that fidgeting traces can reflect both the type and intensity of cognitive load, we next asked whether fidgeting behavior can estimate a participant’s performance in Phase 1 of the PASAT. We framed this as an 11-class classification problem, predicting which performance bin the participant achieved (0-9%, 10-18%, 19-27%, etc., as in figure 2).

We operationalized performance as percent correct in Phase 1, counting non-responses as incorrect. We measured the precision of performance prediction through participant-based 5-fold cross-validation. In each fold, we held back data from a disjoint set of five participants for testing, we split data from the remaining participants 80%/20% for training and validation, and averaged the accuracy of the model’s performance bin predictions across folds. We trained a 1D CNN+LSTM+FCNN architecture against a supervised signal composed of the participant’s Phase I PASAT score paired with a 0.5 second window of fidgeting data as described

above. As shown in Figure 2, the model achieved 79.7% overall accuracy and 79.4% macro-F1 across 11 bins, substantially exceeding the 9% random-guess baseline. These results indicate that fidgeting traces carry quantitative information about task performance in addition to qualitative information about cognitive load levels.

6 Discussion

Our experiments show that spontaneous fidgeting patterns reliably encode information about graded time pressure and task performance during a time-pressured task. Models performed well above chance in detecting levels of PASAT-induced time pressure and in predicting PASAT Phase-1 performance from tactile interaction alone. These results indicate that fidgeting reflects task-specific cognitive load and performance demands, supporting its role as a meaningful behavioral signal during time-pressured cognitive activity such as PASAT.

From an HCI perspective, these findings highlight the potential for integrating tactile behavior into adaptive interfaces, enabling systems to respond dynamically to user cognitive load without explicit interaction. Such applications could support inclusive design, especially for users with ADHD and related conditions.

7 Limitations

This paper reports an initial investigation into the cognitive state information encoded in spontaneous fidgeting behavior, and several limitations should be noted. First, the hardware design of the fidget ball may not fully represent the diversity of tactile affordances that people naturally engage with [5, 17]. Allowing participants to use their own preferred objects, or more ubiquitous devices such as smartphones [3], could enhance ecological validity. In addition, our analyses focus on a single cognitive task (PASAT) and a single device configuration. While this controlled setting enables precise evaluation of task-induced time pressure, future work should validate whether similar decoding performance generalizes to other

time-pressured tasks, everyday fidget objects, and more ecologically diverse contexts. Second, we matched evaluation protocols to each analysis goal using participant-held-out validation. Despite these controls, analysis-specific confounds remain:

Detecting PASAT time pressure. A potential confound is the rhythmic structure of PASAT prompts. We mitigated this by using fixed 0.5 s windows that never span prompts and by omitting recurrent layers, limiting the model’s ability to learn prompt timing rather than cognitive load.

Predicting PASAT performance. We consider three primary potential confounds: class imbalance across performance bins, practice or fatigue within Phase 1, and participant-specific motor signatures. Performance bins were defined to span the full 0–100 percent range rather than to balance sample size, resulting in moderate class imbalance, though we report macro-F1 in addition to accuracy to reduce sensitivity to this effect. Practice or fatigue could correlate with performance level, but Phase 1 was short, data were segmented into uniform 0.5 second windows, and no explicit position features were provided, limiting simple temporal confounds. Participant-based 5-fold cross-validation with disjoint test participants mitigates identity leakage. This strategy does not directly correct class imbalance or within-phase temporal effects, so some residual influence of these factors may remain, though they are unlikely to fully explain the substantially above-chance results.

8 Conclusions and Future Work

This work demonstrates that tactile fidgeting behavior encodes cognitive load and task performance during PASAT. These results suggest that fidgeting provides an informative behavioral signal of task-induced cognitive load and performance demands. This highlights the potential for applications such as cognition-aware technologies that support self-regulation through unobtrusive recognition of cognitive state.

Looking ahead, several directions can strengthen and extend this line of research. Future studies should examine finer grained cognitive load dynamics, validate findings across additional time pressured tasks, and include larger and more diverse participant populations to assess generalizability. Future work should also examine what other cognitive and affective features can be extracted from fidgeting.

At an applied level, smart fidgeting devices could be developed that employ machine learning analysis of tactile signals, such as those presented in this paper, to provide real-time biofeedback for cognitive load or attention regulation, and touch-based behavioral markers may augment clinical evaluations with objective data. Integrating tactile signals with multimodal inputs such as heart rate, posture, or facial expression represents another promising avenue, supported by cross-disciplinary collaboration.

Acknowledgments

This work was supported by the National Institute of Mental Health under Award No. 1R21MH121901. We thank the clinical research team and the study participants for their contributions.

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